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Simple RAG Consumption with Azure OpenAI and AI Search

Informatica Cloud Application Integration Simple RAG Consumption with Azure OpenAI and AI Search
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Preface

Use *Simple RAG Consumption with Azure OpenAI and AI Search* to learn how to receive a query from the user, convert it into vectors, form the context, and return a comprehensive response using a Large Language Model (LLM). This guide assumes that you have an understanding of the Azure OpenAI Connector and Azure AI Search Connector concepts.

CHAPTER 1

Introduction to Simple RAG Consumption with Azure OpenAI and AI Search recipe

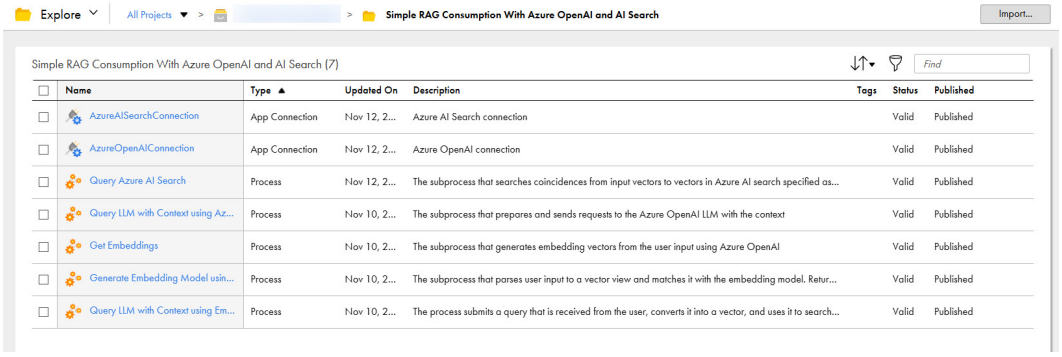
The Simple Retrieval Augmented Generation (RAG) Consumption with Azure OpenAI and AI Search recipe is based on REST and SOAP APIs.

The process submits a query that is received from the user, converts it into a vector, and uses it to search for similar vectors in a database. The top K matches are retrieved, filtered by a cutoff score, and used to form a context. This context including the original query is passed to a Large Language Model (LLM) to generate and return a comprehensive response.

Simple RAG Consumption with Azure OpenAI and AI Search recipe contents

The Simple RAG Consumption with Azure OpenAI and AI Search recipe contains app connections and processes.

The following image shows the assets that the Simple RAG Consumption with Azure OpenAI and AI Search recipe package contains:



Simple RAG Consumption with Azure OpenAI and AI Search recipe assets

The following table lists the assets that the Simple RAG Consumption with Azure OpenAI and AI Search recipe package contains:

Asset Name	Asset Type	Description
AzureAISearchConnection	App connection	Azure AI Search connection.
AzureOpenAIConnection	App connection	Azure Open AI connection.
Query Azure AI Search	Process	Subprocess that searches coincidences from input vectors to vectors in Azure AI search specified as input parameters Index_Host_AzureAISearch and returns a result with metadata.
Query LLM with Context using Azure OpenAI	Process	Subprocess that prepares and sends requests to the Azure OpenAI LLM with the context.
Get Embeddings	Process	Subprocess that generates embedding vectors from the user input using Azure OpenAI.
Generate Embedding Model using Azure OpenAI	Process	Subprocess that parses user input to a vector view and matches it with the embedding model. Returns the matching context score and metadata.
Query LLM with Context using Embeddings Model	Process	Submits a query that is received from the user, converts it into a vector, and uses it to search for similar vectors in a database. The top K matches are retrieved, filtered by a cutoff score, and used to form a context. This context including the original query is passed to a Large Language Model (LLM) to generate and return a comprehensive response.

CHAPTER 2

Using the Simple RAG Consumption with Azure OpenAI and AI Search recipe

To use the Simple RAG Consumption with Azure OpenAI and AI Search recipe, you must perform the following steps manually:

1. Copy and access the recipe
2. Configure and publish the Azure OpenAI connection
3. Configure and publish the Azure AI Search connection
4. Configure and publish the processes
5. Invoke the process

Step 1. Copy and access the recipe

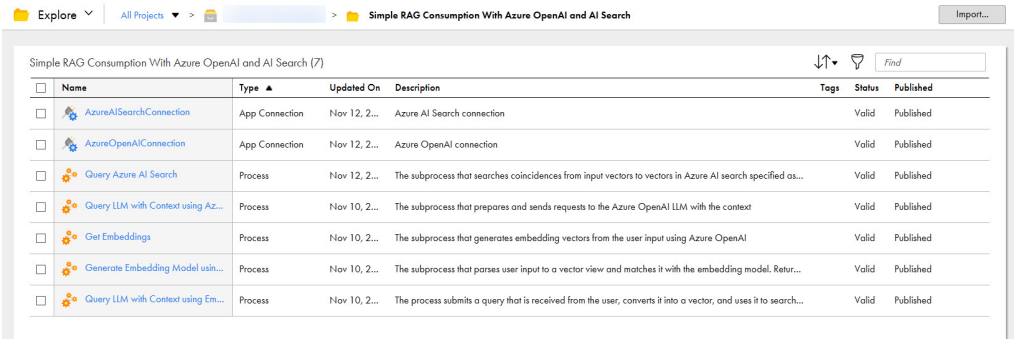
Copy the pre-configured assets in the recipe to a separate project or folder.

1. Open the **Simple RAG Consumption with Azure OpenAI and AI Search** recipe and click **Use**.
2. Select the location where you want to copy the recipe, and then click **Continue**.
3. In the **Copying the recipe** dialog box, click **OK**.

   It might take some time for the recipe to get copied. You will receive a notification when the recipe is ready for use.
4. After the recipe is copied, click **Explore** to access the recipe content.

5. Navigate to the project or folder where you copied the recipe or enter the recipe name in the **Find** box.

   All the assets in the recipe are displayed as shown in the following image:



Step 2. Configure and publish the Azure OpenAI connection

Configure the endpoint URL and API key to connect to the Azure OpenAI connection, and then publish the AzureOpenAIConnection connection.

1. Open the **AzureOpenAIConnection** connection.
2. In the **Connection Properties** section, enter values for the following properties:

Property	Description
Endpoint_URL	The REST API endpoint for Azure OpenAI. You can find this value in the **Keys & Endpoint** section when examining your resource from the Azure portal. Alternatively, you can find the value in **Azure OpenAI Studio** > **Playground** > **Code View**.
API_Key	The API key to authenticate Azure OpenAI connection requests. You can find this value in the **Keys & Endpoint** section when examining your resource from the Azure portal. Alternatively, you can find the value in **Azure OpenAI Studio** > **Playground** > **Code View**.

3. Save and publish the connection.

Step 3. Configure and publish the Azure AI Search connection

Configure the host name and API key to connect to the Azure AI Search connection, and then publish the AzureAISearchConnection connection.

1. Open the **AzureAISearchConnection** connection.

2. In the **Connection Properties** section, enter values for the following properties:

Property	Description
Host_Name	Required. The host name of the search service that you want to connect to. You can find the value in **Microsoft Azure Search Service** > **Overview**, and copy the **Url** field value.
API_Key	Required. The admin API key to authenticate Azure AI Search connection requests. You can find the value in **Microsoft Azure search service** > **Settings** > **Keys**, and copy the primary or secondary admin key.

3. Save and publish the connection.

Step 4. Configure and publish the processes

To configure and publish the processes, perform the following steps:

1. Open the following processes in the order specified below:
    1. Generate Embedding Model using Azure OpenAI
    2. Get Embeddings
    3. Query Azure AI Search
    4. Query LLM with Context using Azure OpenAI
    5. Query LLM with Context using Embeddings Model
2. For each process, on the **Start** tab of the Start step, select **Cloud Server** from the **Run On** list.
3. Optionally, for the **Query LLM with Context using Embeddings Model** process, in the **Set LLM Models** step, in the **Assignments** tab, you can update the values in the **Set_API_Version** and **Set_Embedding_Model** fields.
4. Save and publish all the processes.

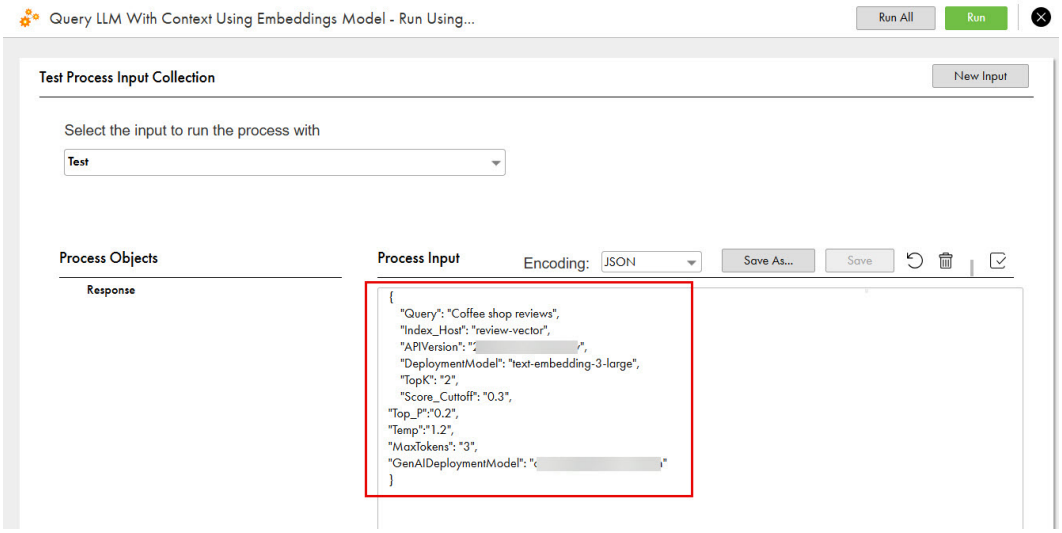
Step 5. Invoke the process

When you invoke the Query LLM with Context using Embeddings Model process, the user sees the answer matching the context that was used in the LLM request.

You can run the process using one of the following options:

- Run Using by passing the input parameters
- REST or SOAP API endpoints in any API client such as cURL, Postman, SOAP UI, or any programming language
- Web browser by passing the input parameters

For example, you can pass the input parameters with the Run Using option as shown in the following image:


Query LLM With Context Using Embeddings Model - Run Using...
Run All
Run
Test Process Input Collection
New Input
Select the input to run the process with
Test
Process Objects
Response
Process Input
Encoding:
JSON
Save As...
Save
{
"Query": "Coffee shop reviews",
"Index_Host": "review-vector",
"APIVersion": "2
/",
"DeploymentModel": "text-embedding-3-large",
"TopK": "2",
"Score_Cuttoff": "0.3",
"Top_P":"0.2",
"Temp":"1.2",
"MaxTokens": "3",
"GenAIDeploymentModel": "
"
}